

1 Introduction

Among the constraints that govern syntactic movement are restrictions on the linear crossing of paths. These restrictions are well-known from the $\bar{A}$ -system of English (Kuno & Robinson, 1972; Culicover & Wexler, 1977; Pesetsky, 1982), where they give rise to the following constraint. It is possible for clauses to host multiple chains of WH-movement if the paths are nested within each other, as in (1a), but it is impossible for clauses to host chains of WH-movement that cross (1b). The same restriction extends to tough-movement, topicalization, and too/enough movement, and I will refer to it as the Crossing Constraint.

(1) *The Crossing Constraint*

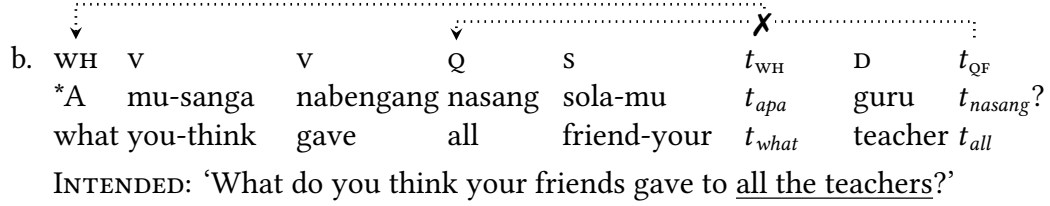
- a. ✓WHAT SUBJECT do you know WHO to talk to ____₂ about ____₁?
- b. ✗WHO do you know WHAT SUBJECT to talk to ____₂ about ____₁?

The standard analysis of the Crossing Constraint, from Pesetsky 1982, runs as follows: (1) the linear string crossed over by an $\bar{A}$ -dependency forms a syntactic object called a PATH, and (2) crossing PATHS are banned in the syntax by a representational constraint. This analysis operates in a precise and intuitive way, but there are many ways in which it fails to fit neatly into the current theoretical context. The notion of PATH has no obvious status in the Minimalist framework, and its position becomes doubly suspicious once linear order is removed from the syntax (Chomsky, 1995). Representational constraints on containment, in turn, have no clear position in a derivational theory of syntax where the trigger for movement (AGREE) operates at a stage without linear order.

The goal of this paper is thus to revisit the theory of paths through a study of Mandar (Austronesian; Indonesia). Many types of movement in this language obey the Crossing Constraint, and the following example provides an example: when WH-movement targets the object in a vso clause, it is possible for quantifier float to operate inside its path and draw the quantifier *all* out of the subject to follow the verb (2a). But it is ungrammatical for quantifier float to cross the path of WH-movement, ruling out examples like (2b).

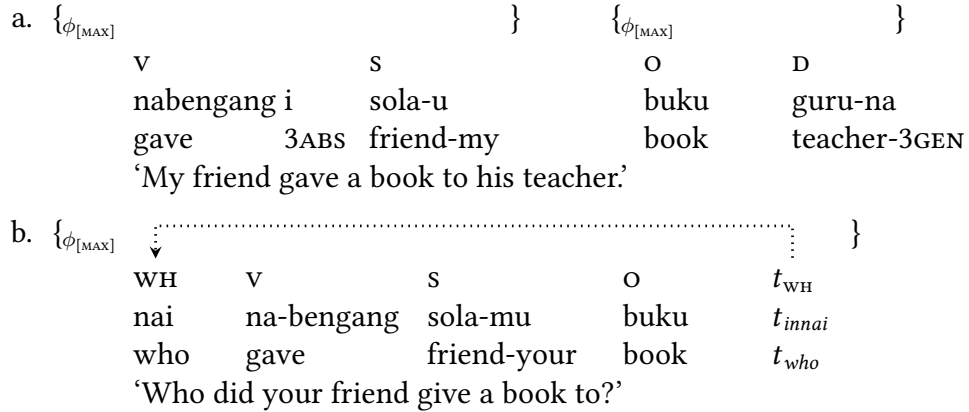
(2) *Crossing Constraints in Mandar*

- a. WH V V Q S t_{QF} t_{WH} D
A mu-sanga nabengang nasang sola-mu t_{nasang} t_{apa} guru?
what you-think gave all friend-your t_{all} t_{what} teacher
‘What do you think all your friends gave to the teachers?’



The first task of the paper is to show that these Crossing Constraints are bound up with a generalization about prosody. In Mandar, the movement operations that cannot cross have a consistent phonological signature: their paths must be parsed into prosodic constituents. More specifically, I will argue that the paths of these specific movements are forcibly parsed into single phonological phrases by a relativized constraint: WRAP(CHAIN). This following diagram shows this effect: the string $vsoD$ (D = applied argument) is normally parsed into two maximal phonological phrases (3a), but when the D undergoes WH-movement, the full string $WH-vso-t$ forms a single maximal phonological phrase (3b).

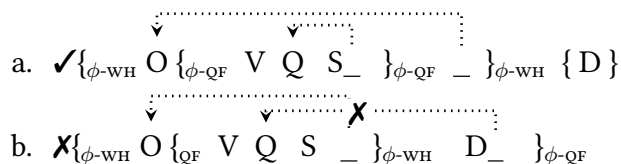
(3) *Certain Paths $\rightarrow$ Prosodic Constituents*



The second task of the paper is then to develop an analysis of the Crossing Condition that is grounded in prosodic constituency. Our starting prosodic generalization suggests that when paths nest, two phonological phrases will be built in a nested configuration (4a). This hypothetical configuration is one that shows prosodic recursion—a phenomenon that has been decisively established to be possible at the level of the phonological phrase (Ladd 1986; Wagner 2005, 2010; Elfner 2012, 2015). Our prosodic generalization leads us to predict further that crossing paths should be parsed into phonological phrases that intersect (4b). This hypothetical configuration is one that will violate a foundational principle of constituency: it is not possible for constituents to intersect (the Proper Bracketing Condition of Nespor & Vogel 1986, in Prosodic Hierarchy Theory). We thus arrive at a correlation between phonological possibilities and linear realities:

prosodic recursion is possible and thus the relevant movement paths can freely nest, but prosodic constituents cannot overlap and thus the relevant movement paths cannot cross.

(4) *Paths and Prosodic Constituency*



The main thrust of the paper is thus to argue that these Crossing Constraints are fundamentally phonological effects. This analysis dispenses with the syntactic notion of PATH and captures the Crossing Constraint in the terminology of prosodic constituency. In the same vein, it sidesteps the need for a representational constraint on paths in the syntax and enforces the relevant restriction in the global evaluation of the phonology. In doing so, it draws together many new phonological generalizations and integrates them with the paradigm of Prosodic Hierarchy Theory. The analysis thus yields a simpler syntax, a refined vision of the syntax-prosody interface, and a better understanding of Mandar.

The remainder of this paper is organized as follows. Section 2 provides background information on Mandar, introduces the prosodic hierarchy, provides diagnostics for the maximal phonological phrase, and introduces the phonological signature of WH-movement. Section 3 presents the Crossing Constraint and shows that it correlates with phonological generalizations; Section 4 develops the analysis. Section 5 concludes with a reflection on the consequences for Match Theory and for syntactic theory at large.

2 Background

Mandar is an Austronesian language that is spoken by 400,000 people in West Sulawesi, Indonesia. It is a member of the South Sulawesi subfamily and has been the subject of an Indonesian-language grammar (Pelenkahu *et al.*, 1983), a description of adverbs (Sikki *et al.*, 1987), a compilation of traditional poetry (Muthalib & Sangi, 1991), and a conversational handbook (Friberg & Jerniati, 2000). The language can also be compared to the other languages of the South Sulawesi subfamily, which have been described in a larger body of work in English (Friberg, 1991, 1996; Strømme, 1994; Matti, 1994; Valkama, 1995a,b; Jukes, 2006; Kaufman, 2008; Laskowske, 2016; Finer, 1997, 1999; Béjar, 1999). This paper will focus on the standard variety, spoken on the coast between the cities of Polewali and

Majene.¹

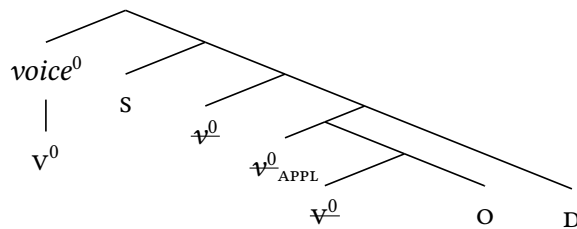
Mandar shows a consistent basic word order of v-s-o-D-x (Brodkin, To Appear) (verb > external argument > internal argument > applied argument > VP-level adjuncts). The language allows *pro*-drop and shows no morphological case-marking. Transitive verbs host ergative agreement with the external argument, and every finite clause contains a second-position clitic that spells out agreement with the absolutive argument. A Mandar clause is presented below, showing the word order and the system of agreement.

(5) *The Shape of a Mandar Clause*

U-solangang i yau iKaco' mua' lamba i *pro* manini.
1ERG-accompany 3ABS 1SG NAME if go 3ABS later
'I'll accompany Kacho' if he goes later.'

Brodkin To Appear argues that the vsod word order of the language is derived from word-building head-movement of the verb (Harizanov & Gribanova, 2019), placing the v in *voice*⁰ (Collins, 2005; Merchant, 2013). This process yields vsod order in tandem with two further points: the s, o, and D must usually be spelled out in their base positions, and the D must be introduced in a rightward specifier of *appl*⁰, yielding the VP-syntax in (6). The following investigation will proceed in rough independence of many systems in the language beyond this point, including the Western Austronesian voice system and the particular systems of agreement and anti-agreement.

(6) *The Derivation of VSOD Order in Mandar*



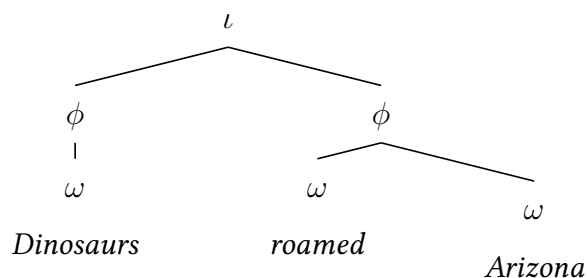
2.1 Prosodic Background

The theory of the Prosodic Hierarchy holds that phonological strings are also organized into hierarchical constituent structures above the metrical level of syllables and feet (Selkirk,

¹GLOSSING: ABS: absolutive, ANTIP: antipassive, APPL: applicative, ERG: ergative, GEN: genitive; 1SG: first-person singular. ORTHOGRAPHY: <c>: /tʃ/, <'>: /ʔ/, <ng>: /ŋ/.

1978, 1984, 1986; Nespor & Vogel, 1986). In the minimal and universal framework of Itô & Mester 2007, this suprametrical constituent structure is built from three kinds of abstract phonological constituent: the prosodic word (ω), the phonological phrase (ϕ), and the intonational phrase (ι). An example of this structure is shown in diagram (7), which reflects the parse of the clause *Dinosaurs roamed Arizona*. There, the subject, verb, and object each form ω s and the verb and object are parsed into a single ϕ .

(7) *Phonological Organization above the Word*



The following investigation will investigate the distribution of the intermediate constituent on this suprametrical hierarchy: the phonological phrase. To do so, we will leverage a set of phonotactic restrictions in Mandarin that interact in crisp and categorical ways to reveal the distribution of the largest phonological phrases in each intonational phrase of the language (Brodкин, To Appear). These restrictions are easy to hear in the spoken language, and they are typically reflected in the popular orthography, but they are often enforced imperfectly in regular speech—as both patterns of phonological phrasing and processes in the phrasal phonology can be derailed by the phonology of focus and by a wide range of non-phonological factors, including speech rate and planning constraints. As a result, our investigation will focus on the ways in which speakers judge these restrictions to emerge in well-planned clauses that are produced at a regular speech rate under broad focus. The focus of our investigation will thus fall on the prosody that is judged to be optimal in this context, setting aside many delicate questions for future work.²

²The judgments in this study have been gathered over six months of targeted work with one speaker of Mandarin, Jupri Talib, with whom I have been working for five years overall (and for three years on prosodic phonology). The phonotactic diagnostics that reveal the distribution of maximal phonological phrases were first noted through natural observation. Their distribution was documented slowly over time, as the influence of focus was sorted out, Jupri was taught the essentials of prosodic hierarchy theory, and a strategy was developed to study prosody through pairwise comparison. To collect each judgment, Jupri was given a pair of written sentences that differed in the application of a phonological rule. Once the sentences were established as pre-phonologically acceptable, he was then walked through a context in which the sentences

2.2 Detecting Prosodic Constituents

We begin with the building blocks of high-level prosodic structure. In Mandar, the prosodic word (ω) can be detected from the distribution of word-level stress, which occurs once per word and falls on the penult (marked phonetically with a low tone; Pelenkahu *et al.* 1983).³ Above the ω , we can detect the distribution of phonological phrases (ϕ s) from the distribution of a high tone (H), which falls at the right edge of most xPs (Brodkin 2025). The following example shows the distribution of stress and the H tone in a vsx clause.

- (8) *Diagnostics for the Word and Phrase*
- | | | | | |
|-------------------------|------|------|----------------------------|----------------|
| máne wémme ^H | i | iti | āndʒóro káiaŋ ^H | ó ^H |
| mane bemme | i | itiŋ | andʒoro káiaŋ ^H | ó ^H |
| just fall | 3ABS | that | coconut big | there |
- ‘That big coconut there just fell.

In this example, the ADVERB-VERB string *mane wemme* “just fell” carries two word-level stresses and a single H tone, revealing that it forms a ϕ that contains two ω s. The following DEM-NOUN-ADJ string *iting anjoro kaiyang* “that big coconut” also carries two word-level stresses and a single H, suggesting the same parse. The x, finally, carries one word-level stress and one H, suggesting that it forms a ω that is contained within a ϕ .

Our investigation will focus on the ways in which minimal ϕ s are organized into a higher level of prosodic constituency. This higher level of constituency can be detected through phonotactic restrictions in the phrasal phonology, and the following section presents four. First, Mandar has a process of Coalescence that reduces /ai ae/ to [e] and /au ao/ to [o]. The following example shows how this process operates in three-word vsx clauses. When the verb ends in a vowel sequence in this context, it must show coalescence under default prosody (/naitai/ “seek” → [néte]). But when the s ends in a final vowel sequence in this context, it cannot show coalescence under default prosody (/balao/ “mouse” → [wálo]). The same restriction holds over the x (/di bangae/ → [di wáŋge]).

could be uttered under broad focus. After that, he was asked to produce each sentence several times. Once the sentences had become familiar, Jupri was then asked whether one version could be identified as acceptable and comparatively “more normal” (Indonesian: *lebih biasa*) in contexts that did not place narrow focus on any constituent (*konteks biasa*). He was then asked to rate the second example as “also possible with regular prosody” (*bisa juga dengan irama biasa*), “only possible with narrow focus” (*hanya dengan penekanan khusus*), or “incorrect” (*salah*). The languages of contact in this work were Indonesian and Mandar.

³Word-level stress falls on the lexical heads N⁰, V⁰, ADJ⁰, and ADV⁰ but not the functional x⁰s P⁰, C⁰, POL⁰, and ASP⁰ (which hosts auxiliaries) in Mandar. Brodkin 2025 argues that the distribution of word-level stress arises from the right-alignment of a single trochee in the ω .

- (9) *Coalescence: The First Asymmetry*
- | | | | |
|-------------------|------|--------------------|------------------------|
| néte ^H | i | waláo ^H | di wangáe ^H |
| na-itai | i | balao | di bangae |
| 3ERG-look for | 3ABS | mouse | in PLACE |
- ‘The mouse looked for it in Banggae.’

Second, Mandar has a process of Gliding that reduces prevocalic /i u/ to [j w]. In three-word vsx clauses, it applies at the right edge of the v but not at the right edges of the s or x. Thus in example (??), the v /napapia/ “make” reduces to [napápja], but the s /imaria/ “NAME” cannot reduce to [imárja] and the x /lamasarian/ cannot reduce to [lamasárjan].

- (10) *Gliding: A Second Asymmetry*
- | | | | |
|----------------------|------|---------------------|---------------------------|
| napápja ^H | i | imaría ^H | di lamasaría ^H |
| na-papia | i | imaria | di lamasarian |
| 3ERG-make | 3ABS | NAME | in PLACE |
- ‘Maria made it in Lamasariang.’

THIRD: Mandar has a process that deletes intervocalic /ʔ/ (which is always parsed as a coda) and intervocalic coda /ŋ/ (which is parsed as a coda word-finally). I will refer to this process as Glottal Coda Deletion, building from the understanding (well-motivated in Mandar, I believe), that /ŋ/ is phonologically a glottal segment (De Lacy, 2006). This process has the same distribution as Coalescence and Gliding: in a three-word vsx clause, it applies in the v to reduce /nakaraʔus/, “scratch” to [nakáros] (with Glottal Coda Deletion feeding Coalescence). But it is blocked at the right edge of the s, /iripaʔi/ (a NAME), which cannot reduce to [irípe], and the x /rebataʔa/, which cannot reduce to [rewáta].

- (11) *Glottal Coda Deletion: A Third Asymmetry*
- | | | | |
|----------------------|------|----------------------|--------------------------|
| nakáros ^H | i | iripáʔi ^H | di rewatáʔa ^H |
| na-karaʔus | i | iripaʔi | di rebataʔa |
| 3ERG-scratch | 3ABS | NAME | in PLACE |
- ‘Ripa’i scratched it in Rebata’a.’

Fourth, Mandar has a process of Voiced Obstruent Lenition that lenites intervocalic /b d d̥ ɡ/ to [w ɿ j ɣ]. At the junctures that require Coalescence, Gliding, and Glottal Coda Deletion at the right edge of one ϕ, this process applies at the left edge of the next. In a three-word vsx clause, then, Voiced Obstruent Lenition must apply at left edge of the s when the preceding segment is a vowel: thus the s /do guru/ “that teacher” is reduced to [ɿo ɣúru] (??). Voiced Obstruent Lenition is then blocked at the left edge of the x, such that the x /di genaʔ/ “earlier” cannot be realized as [ɿi ɣénaʔ].

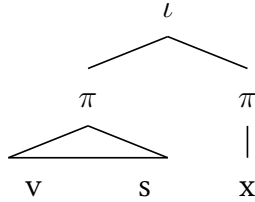
(12) *Voiced Obstruent Lenition: A Fourth Asymmetry*

nawátʃa ^H	i	.io	yúru ^H	di	yénaʔ ^H	ó ^H
na-batʃa	i	do	guru	di	genaʔ	o
3ERG-read 3ABS	that teacher			in earlier		there

‘That teacher there read it earlier.’

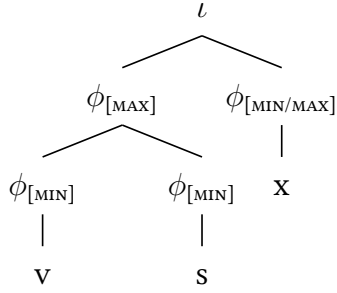
Brodkin To Appear argues that these restrictions reveal the constituency in diagram (13): the vs string forms a constituent that allows Coalescence, Gliding, Glottal Coda Deletion, and Voiced Obstruent Lenition internally. The right edge of this constituent blocks the first three processes, banning final Coalescence, Gliding, and Glottal Coda Deletion in the s and x. The left edge of this constituent blocks the last process, banning initial Voiced Obstruent Lenition in the x. As these restrictions cannot be keyed to syntactic constituents (the hypothesis of Syntax-Free Phonology; **Nespor & Vogel 1986**), we must identify the relevant constituent as an abstract category of the phonology: namely, a prosodic constituent that falls between the minimal ϕ and the intonational phrase (ι).

(13) *The VSX String: Constituency*



There is a natural candidate for a constituent in this position, and it is the maximal ϕ : namely, the type of ϕ that is directly dominated by the ι (**Itô & Mester, 2007, 2012, 2013**). The following diagram illustrates the shape of this analysis. We have already seen that the v, s, and x are each parsed into minimal ϕ s, as they each host the ϕ -final H tone (8). By hypothesis, the minimal ϕ s that contain the v and the s are then parsed into another higher ϕ , yielding the recursive prosodic constituency in (14). This higher ϕ is directly dominated by the ι , rather than being dominated by another ϕ . As a result, it acquires the relational property of maximality, marked with the subscripted [MAX] below. The minimal ϕ that contains the x is also rendered maximal for this reason, as it is not parsed into a higher ϕ with any other constituent. It is the property of maximality that yields the restrictions on Coalescence, Gliding, Glottal Coda Deletion, and Voiced Obstruent Lenition above.

(14) *The Relevant Constituent: The Maximal ϕ*



2.3 Prosody and Syntax

The suprametrical constituent structure of the prosody is grounded in the constituency of the syntax beneath it. The intermediate constituent ϕ , for instance, is widely understood to be built in response to an interface pressure to reflect the distribution of syntactic xps (Selkirk, 1984, 1986; Selkirk & Tateishi, 1991; Selkirk & Shen, 1990; Selkirk, 1995; Nespor & Vogel, 1986; Selkirk, 2009; Elfner, 2012, 2015). Following Itô & Mester 2019, I will assume that this requirement is one of existential correspondence, which demands that every xp in the syntax be placed in correspondence with a ϕ in a global and output-oriented calculus (15a). Each ϕ is then aligned with its corresponding xp by the PROSODY-TO-SYNTAX MATCH constraint in (15b).

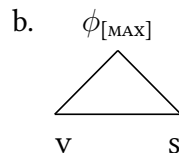
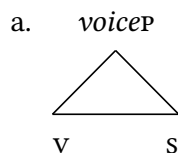
(15) *The Isomorphism Constraints*

- a. MAX-XP: Let sx be an input syntactic representation and pr its corresponding output representation. Assign one violation for every xp in sx which does not correspond to a ϕ in pr.
- b. MATCH(ϕ ,xp): Let sx be an input syntactic representation and pr its corresponding output representation. Assign one violation for every ϕ in pr that has a corresponding xp in sx such that the left and right edges of the ϕ not aligned with the left and right edges of the xp.

These constraints work together to organize surface strings into phonological constituent structures that preserve the constituency of the syntax. In the simplest cases, they force maximal xps to form ϕ s, yielding the regular mapping of complete NPs and PPs to ϕ s in Mandar. In more complex cases, they force the recursive mapping of functional projections to ϕ s as well (Elfner, 2012, 2015; Itô & Mester, 2019). This requirement gives

rise to the mapping of the vsx string: the v and s form a constituent in the syntax (16a), and this higher constituent is mapped to a ϕ to yield the recursive parse above (16b).

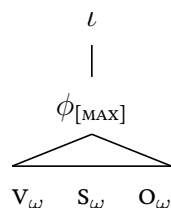
(16) *Regular Mapping: $XP \rightarrow \phi$*



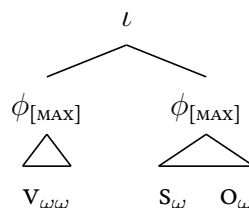
Despite this rough correspondence, the constituent structure of the prosody is known to deviate from the syntax in response to several types of grammatical constraints. Setting aside the phonology of focus (Féry, 2013), there are at least two types of pressures that are held to drive mismatches in the interface paradigm of Match Theory (Selkirk, 2009, 2011). The first are output-oriented constraints that force the the surface prosodic structure to satisfy requirements of phonological well-formedness. In many languages, for instance, constraints on accentual clash force syntactic constituents to be split up and parsed into prosodic constituents that have no analogues in the syntax (Kubozono, 1989; Selkirk & Elordieta, 2010). In Mandar, similar effects are forced by a constraint on weight (Brodkin, To Appear): the maximal ϕ cannot contain more than three ω s. This restriction triggers disruptions of the following shape: an XP will be parsed into a maximal ϕ if it contains three ω s (17a), but it will be split into multiple maximal ϕ s if it contains four or more (17b).

(17) *The Weight Constraint*

a. Three ω s $\rightarrow \phi_{[MAX]}$



b. Four ω s $\rightarrow \phi_{[MAX]}\phi_{[MAX]}$



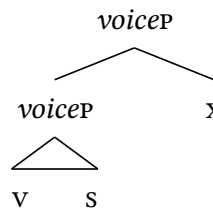
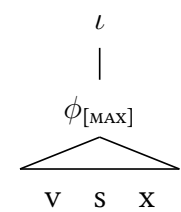
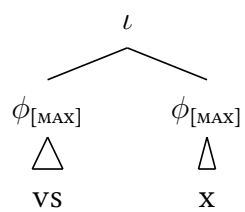
The following examples show this effect. The vso string typically forms a maximal ϕ when the v, s, and o are each one word long (18a). But this string is split into two maximal ϕ s when an additional ω is added anywhere within that span. This effect can be forced by adding a ω -sized prefix to the verb, like the reduplicant (18b). In that case, our phonotactic diagnostics reveal the shift: between the v and the s, Voiced Obstruent Lenition applies when the string is three ω s long (18a) but not when the verbal ϕ contains two ω s (18b).

(18) *The Weight Constraint: Examples*

- a. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
na.íundu^H i yúru^H wálo?^H djoŋiŋ^H
na-dundu i guru ballo? dionjɪŋ
3ERG-drink 3ABS teacher palm wine yesterday
‘The teacher drank the palm wine yesterday.’
- b. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
ná.ɪu-na.íundu^H i gúru^H wálo?^H djoŋiŋ^H
σσ-na-dundu i guru ballo? dionjɪŋ
RED-3ERG-drink 3ABS teacher palm wine yesterday
‘The teacher drank a bit of the palm wine yesterday.’

The second set of pressures that drive mismatches in constituency are those that govern the phonologization of particular types of syntactic information. These pressures are most familiar from the literature on adjunction, which has long been recognized to have a regular phonological reflex (Truckenbrodt 1999). The core effect has already appeared in the data that we have seen so far. In Mandar, the locative and temporal adjuncts in the vsx string are presumably right-adjoined to a constituent like the *voiceP* (19a). If this constituent is dominated by any higher xp, like a higher segment of the *voiceP*, then the full vsx string should be parsed into a higher ϕ when weight constraints allow (19b). But this is not what occurs: instead, every x is parsed into its own maximal ϕ (19c), yielding a mismatch in constituency.

(19) *The Parse of Adjuncts*

- a. Starting Syntax
- 
- b. Expected Parse
- 
- c. Attested Parse
- 

To capture this type of mismatch, Selkirk (2011, 483, fn. 38) proposes that phrase-level adjunction derails the mapping from xps to ϕ s, yielding a version of Match Theory which forces mismatch between syntactic and prosodic constituency to preserve the syntactic distinction between arguments and adjuncts in the prosody. In the same vein, the larger literature has argued that many other syntactic relationships are phonologized in

ways that break the symmetry between syntactic and prosodic constituency, including c-command (Kalivoda, 2018), selection (Clemens & Coon, 2018), and the relationship that links in-situ WH-phrases with c⁰s in various dialects of Japanese (Deguchi & Kitagawa, 2002; Ishihara, 2002, 2003, 2005; Hirotsu, 2005; Smith, 2005, 2013; Richards, 2010, 2016), which is distinct from the simple prosody of focus in at least the Fukuoka dialect (Smith, 2014).

These observations collectively suggest that the optimal constituent structure of the prosody emerges from the interaction of three specific types of pressure at the interface of syntax and phonology: the requirement to mirror the input constituency of the syntax, the requirement to satisfy output-oriented phonological constraints, and the requirement to phonologize fine-grained properties of the syntactic input in specific ways. It is against this backdrop that we will advance on the phonology of WH-movement in Mandarin.

2.4 The Phonology of Mandarin Wh-Movement

Mandarin has a process of WH-movement that obligatorily targets interrogative WH-words and places them in the left periphery. In the following example, this process targets the WH-word *nai* “who” and raises it from the position of the s in a vs clause.^{4,5}

(20) Mandarin Wh-Movement

$\{\phi_{[\text{MAX}]}$			}
ne	yu ^H mó ^H ra ^H	_____	
nai	gumora	t_{nai}	
who	shout	t_{who}	
‘Who’s shouting?’			

⁴The WH-words *a* “what,” *na* “which,” and *nai* “who” trigger another phonological process which shows that they are non-final in the maximal ϕ . These WH-words always form phonological phrases, but like many other x⁰s in the language, they are segmentally too small to satisfy a word-minimality constraint that is enforced in this position: the final ω in the ϕ must be disyllabic. The optimal repair to this constraint, which is favored for all monosyllabic x⁰s in this position, is shown in (20) and all further examples: the ω that corresponds to the WH-word is made disyllabic by pulling in first syllable of the following XP. This process shifts the ϕ -final H-tone one syllable to the right, and it is only possible when the two constituents are parsed into a single maximal ϕ . The application of this process thus reveals that these WH-words are parsed into maximal ϕ s with what follows; see Brodtkin 2025 for documentation and analysis.

⁵Clause-initial WH words are syntactically the predicates of pseudoclefts in Indonesian, the primary language of contact (Cole et al. 2005; see also Potsdam 2009). But these WH-questions in Mandarin involve direct WH-movement: the initial WH-words are demonstrably not predicates and the following constituents are demonstrably not headless relatives (Brodtkin, 2020). Mandarin also has a strategy of pseudoclefting that involves a visibly copular syntax; those constructions are prosodically distinct and are set aside below.

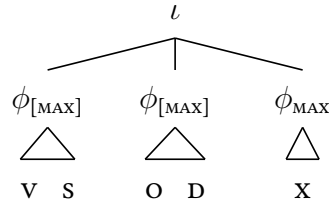
The phonological signature of WH-movement becomes visible when WH-movement crosses strings that contain more than three ω s. The following examples show this effect. Example (21a) presents the baseline: a four-word clause of the order VSOD. This clause is parsed $\{\phi_{[MAX]} \text{ VS} \} \{\phi_{[MAX]} \text{ OD} \}$, as can be seen from our phonotactic diagnostics for the maximal ϕ : Glottal Coda Deletion is blocked at the right edge of the s and Voiced Obstruent Lenition is blocked at the left edge of the o. Example (21b) then shows what happens when WH-movement targets the final D: Coalescence applies at the right edge of the D, Glottal Coda Deletion applies at the right edges of the v and the s, and Voiced Obstruent Lenition applies at the left edge of the o. These effects suggest the parse $\{\phi_{[MAX]} \text{ WH}_D\text{-VSO-}t_{\text{WH}} \}$.

(21) *The Prosodic Effect of Wh-Movement*

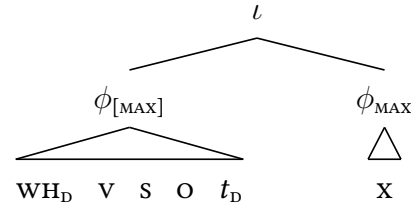
- a. $\{\phi_{[MAX]} \quad \quad \quad \}$ $\{\phi_{[MAX]} \quad \quad \quad \}$ $\{\phi_{[MAX]} \quad \}$
 nawéŋa^H ĩ iripáʔi^H do taláye^H yúru^H ó^H
 na-beŋaŋ i iripaʔi do talagae guru o
 3ERG-give 3ABS NAME that tomato teacher there
 ‘Ripa’i gave the teacher that tomato there.’
- b. $\{\phi_{[MAX]} \quad \quad \quad \}$ $\{\phi_{[MAX]} \quad \}$
 né na^Hwéŋa^H ĩrípe^H .lo talayáe^H — ó^H
 nai na-beŋaŋ NAME do talagae t_{innai} o
 who 3ERG-give NAME that tomato t_{who} there
 ‘Who did Ripa’i give that tomato there to?’

The following diagram shows the basic effect: the string VSOD is typically parsed into multiple maximal ϕ s, yielding the parse $\{\phi_{[MAX]} \text{ VS} \} \{\phi_{[MAX]} \text{ OD} \}$ (22a), but when WH-movement targets the D, the string WH_D-VSO- t_{WH} is parsed into a singular maximal ϕ (22b).

(22) a. *Without WhM*



b. *WhM of the D*



When we turn to other cases of WH-movement, we can see that this initial shift fits into a larger generalization. The following example, for instance, shows the parse that emerges when WH-movement targets the o in a clause of the order VSOD: there, the parse shifts from $\{\phi_{[MAX]} \text{ VS} \} \{\phi_{[MAX]} \text{ OD} \}$ to $\{\phi_{[MAX]} \text{ WH}_O\text{-VS-}t_{\text{WH}} \} \{\phi_{[MAX]} \text{ D} \}$.

(23) *The Path of Wh-Movement* $\rightarrow \phi$

$\{\phi_{[MAX]}$					$\}$	$\{\phi_{[MAX]}$	$\}$	$\{\phi_{[MAX]}$	$\}$
né	na ^H kiríja ^H ĭ	.io	ripáʔi ^H	—		gúru^H	ó ^H		
nai	na-kiriŋaŋ i	do	NAME	t_{nai}		guru	o		
who	3ERG-send 3ACC	that	NAME	t_{what}		teacher	there		

‘Who did that Ripa’i there send to the teacher?’

These patterns suggest that WH-movement does not force the erasure of all maximal ϕ boundaries in the space after the WH-word. Instead, it forces a prosodic shift in the linear string that it crosses: a ϕ must be built from the surface position of the WH-word to the base position of its trace. I will refer to this effect as the Wrapping Effect.

The Wrapping Effect holds over all cases of clause-internal WH-movement in the language, and it extends equally to cases where WH-movement crosses out of an embedded clause. The following examples illustrate. Example (24a) presents a baseline clause where the matrix verb *hara* ‘hope’ selects a complement CP. This clause is parsed $\{\phi_{[MAX]} \text{ VS} \} \{\phi_{[MAX]} \text{ VS} \}$, as can be seen from our phonotactic diagnostics for the maximal ϕ . Example (24b) then shows what happens when WH-movement targets the embedded s: Glottal Coda Deletion applies at the right edge of the matrix s and Voiced Obstruent Lenition applies at the left edge of the embedded v. These effects suggest the parse $\{\phi_{[MAX]} \text{ WH}_s\text{-VSV-}t_{\text{WH}} \}$.

(24) *Long-Distance Wh-Movement* $\rightarrow$ Same Effect

- a. $\{\phi_{[MAX]}$
- | | | | | | | | | | |
|---------------------------|------|------|---------------------|---------------------------|------|-------------------|----------------|------------------|------|
| | | | | | $\}$ | $\{\phi_{[MAX]}$ | $\}$ | $\{\phi_{[MAX]}$ | $\}$ |
| nahára^H | i | .io | ripáʔi ^H | gumóra^H | i | yúru ^H | ó ^H | | |
| na-haraʔ | i | do | ripaʔi | gumora | i | guru | o | | |
| 3ERG-hope | 3ABS | that | NAME | scream | 3ABS | teacher | there | | |
- ‘That Ripa’i there hopes that the teacher will scream.’
- b. $\{\phi_{[MAX]}$
- | | | | | | | | | | |
|-----------|-----------------------------------|------|-------------------|---------------------|-----------|------------------|----------------|--|--|
| | | | | | $\}$ | $\{\phi_{[MAX]}$ | $\}$ | | |
| né | na ^H hára ^H | .io | rípe ^H | yumóra ^H | — | | ó ^H | | |
| nai | na-haraʔ | do | ripaʔi | gumora | t_{nai} | | o | | |
| who | 3ERG-hope | that | NAME | scream | t_{who} | | there | | |
- ‘Who does that Ripa’i there hope will scream?’

The Wrapping Effect extends equally to embedded questions, where WH-phrases only raise to the left periphery of the embedded clause. The following examples illustrate. Example (25a) presents the baseline, in which the matrix verb *issang* ‘know’ selects a complement CP of the order VSOD. This clause is parsed $\{\phi_{[MAX]} \text{ VS} \} \{\phi_{[MAX]} \text{ VS} \} \{\phi_{[MAX]} \text{ OD} \}$. Example (25b) then shows what happens when WH-movement operates in this embedded

clause and targets the D: the prosody of the matrix clause is undisturbed and the prosody of the embedded clause is reorganized, yielding the parse $\{\phi_{[\text{MAX}]}\text{ VS}\}\{\phi_{[\text{MAX}]}\text{ WH}_D\text{-VSO-}t_{\text{WH}}\}$.⁶

(25) *Embedded Wh-Movement → Same Effect*

- a. $\{\phi_{[\text{MAX}]}\}$ $\{\phi_{[\text{MAX}]}\}$ $\{\phi_{[\text{MAX}]}\}$
néssa^H ĭ iripá?i^H mwa? naweŋa^H ĭ jáli^H búku^H yúru^H
na-issan ĭ iripa?i mua? na-beŋan ĭ iali buku guru
3ERG-know 3ABS NAME C 3ERG-give 3ABS NAME book teacher
‘Ripa’i knows that Ali gave the teacher a book.’
- b. $\{\phi_{[\text{MAX}]}\}$ $\{\phi_{[\text{MAX}]}\}$
néssa^H ĭ iripá?i^H né na^Hwéŋa^H ĭáli^H wúku^H _____
na-issan ĭ iripa?i nai na-beŋan iali buku *t_{nai}*
3ERG-know 3ABS NAME who 3ERG-give NAME book *t_{who}*
‘Ripa’i knows who Ali gave a book to.’

2.5 The Constraint Wrap-Chain

The task of this subsection is thus to integrate the following generalization into a theory of the syntax-phonology interface: the path from a Mandar WH-word to its trace must form a ϕ . To capture this effect, I propose that the paths of WH-movement in Mandar are parsed into prosodic structure in compliance with new a ranked and violable interface constraint. This constraint is one which governs the phonologization of the input syntactic relationship of WH-movement and mandates the construction of a ϕ that stretches from the surface position of a WH-word to the base position of its trace (thus requiring alignment with a null element, in a somewhat surprising way). Borrowing a family name from [Truckenbrodt 1999](#), I propose that the relevant constraint has the shape in (26).

(26) $\text{WRAP}(\text{WH-CHAIN}, \phi)$

Let sx be an input syntactic representation and pr its corresponding output representation. Assign one violation for every WH-chain in sx such that the surface position of the WH-word and the surface position of its lowest trace are not left- and right-aligned with the left and right edges of a corresponding ϕ in pr .

⁶This observation also shows that the prosody of WH-movement is distinct from the prosody of contrastive focus in Mandar. Setting second-occurrence foci aside, Mandar requires contrastive foci to be final in the first maximal ϕ of the ι . This restriction yields two effects: (i) maximal ϕ boundaries must be created after contrastive foci (focus via Alignment; [Féry 2013](#)), and (ii) the span from the left edge of the ι to a contrastive focus must be parsed into a single maximal ϕ (pre-focal dephrasing). The same generalization does not extend to WH-traces, however, which can be preceded by an unlimited number of maximal ϕ s (25b).

The wrapping effect emerges from the ranking of $\text{WRAP}(\text{WH-CHAIN}, \phi)$ over the other constraints that govern the form of prosodic structure. For instance, $\text{WRAP}(\text{WH-CHAIN}, \phi)$ will continue to demand the construction of ϕ s even when WH-movement crosses over more than three ω s—mandating an overheavy maximal ϕ that violate MAXIMAL TERNARITY: the output-oriented constraint that bans maximal ϕ s of more than three ω s ((27a); see *Brodkin To Appear* for discussion). The following tableau illustrates. In a four word vsod clause with WH-movement of the D, MAXIMAL TERNARITY favors the parse $\{\phi_{[\text{MAX}]}\ \text{WH-}\text{v}\}\ \{\phi_{[\text{MAX}]}\ \text{sot}_{\text{WH}}\}$: one on which the string is split into two maximal ϕ s (Candidate a). $\text{WRAP}(\text{WH-CHAIN}, \phi)$, however, demands the parse $\{\phi_{[\text{MAX}]}\ \text{WH-VSO-}t_{\text{WH}}\}$, where the path of WH-movement is wrapped by a single ϕ (Candidate b). The ranking of $\text{WRAP}(\text{WH-CHAIN}, \phi)$ over MAXIMAL TERNARITY guarantees this result.

(27) a. *Maximal Ternarity*

Assign one violation for every maximal ϕ that contains more than three ω s.

b. *Wrap(wh-chain, ϕ) > Maximal Ternarity*

$[\text{CP}\ \text{WH}_D \dots [\text{voiceP}\ \text{V}\ \text{S}\ \text{O}\ t_{\text{WH}}]]$	$\text{WRAP}(\text{WH-CHAIN}, \phi)$	MAXIMAL TERNARITY
a. $\{\phi_{[\text{MAX}]}\ \text{WH}\ \text{v}\}\ \{\phi_{[\text{MAX}]}\ \text{S}\ \text{O}\ t_{\text{WH}}\}$	*!	
 b. $\{\phi_{[\text{MAX}]}\ \text{WH}\ \text{v}\ \text{S}\ \text{O}\ t_{\text{WH}}\}$		*

There are also cases where $\text{WRAP}(\text{WH-CHAIN}, \phi)$ will force violations of $\text{MATCH}(\phi, \text{XP})$ —the constraint that enforces syntax-prosody isomorphism. This effect can be seen when WH-movement proceeds from the middle of an embedded CP: for instance, when it targets the O in the embedded vsod clause in (23). The following tableau shows the result. $\text{MATCH}(\phi, \text{XP})$ favors parses like $\{\phi_{[\text{MAX}]}\ \text{WH}_O\text{-VS-}t_{\text{WH}}\ \text{D}\}$, where the embedded CP is well-aligned with a ϕ (Candidate a). $\text{WRAP}(\text{WH-CHAIN}, \phi)$, however, forces violations: it favors the parse $\{\phi_{[\text{MAX}]}\ \text{WH}_O\text{-VS-}t_{\text{WH}}\}\ \{\phi_{[\text{MAX}]}\ \text{D}\}$, where the WH-chain forms a ϕ instead (Candidate b).

(28) *Wrap(WH-CHAIN, ϕ) > Maximal Ternarity > Match(ϕ , XP)*

$[\text{CP}\ \text{WH}_O \dots [\text{voiceP}\ \text{V}\ \text{S}\ t_{\text{WH}}\ \text{D}]]$	WRAP	MAXIMAL TERNARITY	$\text{MATCH}(\phi, \text{XP})$
a. $\{\phi_{[\text{MAX}]}\ \{\phi\ \text{WH}\ \text{v}\ \text{S}\ t_{\text{WH}}\}\ \text{D}\}$		*!	
 b. $\{\phi_{[\text{MAX}]}\ \text{WH}\ \text{v}\ \text{S}\ t_{\text{WH}}\}\ \{\phi_{[\text{MAX}]}\ \text{D}\}$			*

3 Path Containment in the Prosody

We thus turn to the Crossing Constraint with the following result in place: there is a phonological reflex to WH-movement in Mandar, and it is not a simple process that operates over a verb adjacent to the WH-word or its trace (like the phonological reflexes

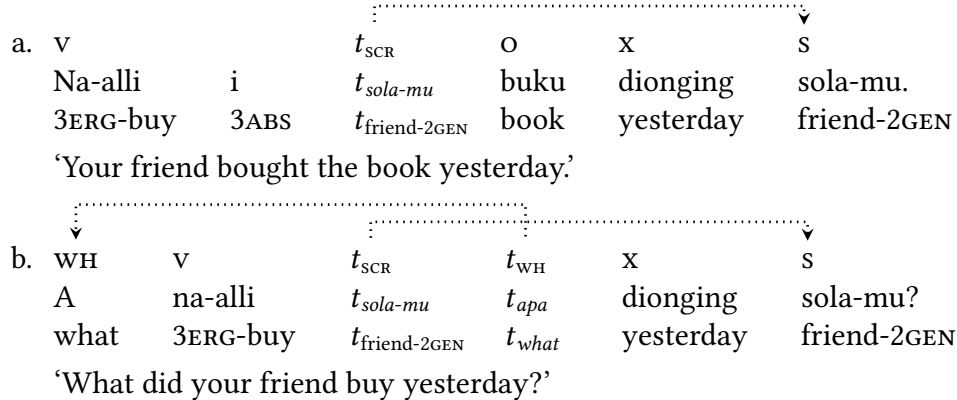
of WH-movement argued to exist in Kikuyu (Clements *et al.*, 1983; Clements, 1984b,a), Igbo (Amaechi & Georgi, 2019), Asante Twi (Korsah & Murphy, 2020) and Welsh (Borsley *et al.*, 2007)). In this language, rather, WH-movement forces a large-scale reorganization of the surface constituent structure of the phonology—and in particular, forces its path to become a tight prosodic domain within that representation.

3.1 Licit Crossings

The classical cases of crossing effects emerge fully within the $\bar{A}$ -system: Pesetsky 1982, for instance, notes that the constraint emerges around WH-movement, relativization, tough-movement, enough-movement, and adverbial topicalization in English. In Mandar, however, the situation is somewhat distinct. In this language, WH-movement shows a range of properties, familiar from many other Western Austronesian languages (Aldridge, 2004), that should independently problematize the derivations where WH-movements cross paths. But this process freely crosses paths with two well-studied $\bar{A}$ -operations, and we will briefly review these facts and explain them below.

Mandar has a process of $\bar{A}$ -scrambling that places definite arguments in a position where they follow right-adjuncts to the *voiceP* (Brodkin, To Appear). Example (29a) shows that this process can move the s to the right to yield the surface order v_ODX-s; example (29b) shows that this step can freely cross the path of WH-movement.

(29) *Scrambling: No Interaction with Wh-Movement*



The same pattern of crossing emerges around topicalization, whose essential effect is shown in example (30a): it places its targets in left-peripheral positions that are prosodically set off from the rest of the clause. The crossing behavior of this process is illustrated in example (30b): it can freely cross the path of WH-movement.

(30) *Topicalization: No Crossing Constraint*

- a. TOP V t_{TOP}
 Do gaja o, na-parakke' $t_{do\ gaja\ o}$ iAli
 that elephant there 3ERG-frighten $t_{that\ elephant\ there}$ NAME
 'That elephant there, it frightened Ali.'
- b. TOP WH V t_{TOP} t_{WH}
 Do gaja o nai na-parakke' $t_{do\ gaja\ o}$ $t_{nai?}$
 that elephant there who 3ERG-frighten $t_{that\ elephant\ there}$ t_{who}
 'That elephant there, who did it frighten?'

Despite the possibility for crossing, it is very clear that both scrambling and topicalization involve genuine syntactic movement. Each operation obeys all of the known island constraints in the language, and as a result, for instance, Topicalization cannot target constituents that originate in clauses embedded in complex NPs (31b): in other words, it obeys the Complex NP Constraint (Ross, 1967). Scrambling, in the same vein, respects both this constraint and the Right Roof Constraint (Ross 1967; on the Right Roof Constraint in Mandarin, see Brodtkin 2024).

(31) *Topicalization and Scrambling: Island Constraints*

- a. U-irrangi i [_{DP} karewa [_{CP} mua' pole i iAli]].
 1ERG-hear 3ABS news that come 3ABS NAME
 'I heard the news that Ali came.'
- b. *iAli, u-irrangi i [_{DP} karewa [_{CP} mua' pole i ____]].
 NAME, 1ERG-hear 3ABS news that come 3ABS
 INTENDED: 'Ali, I heard the news that ____ came.'

Topicalization and scrambling can also target constituents that must reconstruct for the purposes of binding. For instance, Brodtkin & Royer 2024 show that Mandarin has a reflexive anaphor *alawe* that strictly obeys Condition A of the Binding Theory (Reinhart, 1983) (32a). This element can be topicalized (32b) and scrambled (32c).

(32) *Scrambling and Topicalization: Targets can Reconstruct*

- a. Na-pokannyang i sola-na iCicci' **alawe-na** dionging.
 3ERG-trust 3ABS friend-3GEN NAME self-3GEN yesterday
 'Chichi's_i friend_j trusted herself _{*i,j} yesterday.'

- b. Alawe-na, na-pokannyang i sola-na iCicci' _____ dionging.
 self-3GEN 3ERG-trust 3ABS friend-3GEN NAME yesterday
 'Herself, Chichi's friend trusted yesterday.'
- c. Na-pokannyang i sola-na iCicci' _____ dionging alawe-na.
 3ERG-trust 3ABS friend-3GEN NAME yesterday self-3GEN
 'Chichi's friend trusted herself yesterday.'

Brodkin To Appear shows further that the targets of scrambling must reconstruct for the purposes of Condition C. The following examples illustrate. In ditransitive clauses, the o cannot contain an R-expression that is coindexed with a pronoun in the position of the s (33a). This restriction persists when the o is scrambled to the right (33b).

(33) *Scrambling: Reconstruction for Condition C*

- a. Na-na-bengang o [s pro] [o_{ACC} buku [rc na-alli iNina]] marondong.
 will-3ERG-give 2ABS pro book 3ERG-buy NAME tomorrow
 'Tomorrow she_{*i,j} will give you the book that Nina_i bought.'
- b. Na-na-bengang o [s pro] _____ marondong [o_{ACC} buku [rc na-alli iNina]].
 will-3ERG-give 2ABS pro tomorrow book 3ERG-buy NAME
 'Tomorrow she_{*i,j} will give you the book that Nina_i bought.'

We thus conclude that there are two steps of $\bar{A}$ -movement in the language that freely cross paths with WH-movement. This possibility, in turn, correlates neatly with a fact of phonology: the paths of scrambling and topicalization are never wrapped by ϕ s. **Brodkin 2024** shows that the targets of topicalization always form independent intonational phrases (34a).⁷ **Brodkin To Appear**, in turn, shows that the targets of scrambling are systematically parsed into their own maximal ϕ s: for instance, in a three- ω vos clause where the s is scrambled to the right, the parse must be $\{\phi_{[MAX]} \text{ VO} \} \{\phi_{[MAX]} \text{ s} \}$ (34b).

(34) *Topicalization and Scrambling: Path $\nrightarrow \phi$*

⁷The intonational phrase (ι) can be detected through a restriction on a phonological process that applies across the edges of the maximal ϕ . In Mandarin, coda /ŋ/ assimilates to all following segments and denasalizes before all non-nasals but /b d d̥ ʒ g/, applying across ω - and ϕ -boundaries. In example (34b), then, it applies across the maximal ϕ boundary that falls between the s *Ramang* "NAME" and *sola bainena* "with his wife." But it is blocked between the initial external topic *iting bulang* "that month" and the following v *pole* "come," suggesting that the initial external topic forms its own ι . It is impossible for ϕ s to be built around constituents that are parsed into separate ι s, due to the principle of (Weak) Layering (Nespor & Vogel, 1986; Itô & Mester, 1992), and as a result, the path of topicalization cannot be wrapped by a ϕ .

- a. $\{\iota\}$ $\{\iota\}$ $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
- itim búlaŋ^H, póle^H i irámas^H sola wenéna^H _____
- itiŋ bulaŋ pole i iramaŋ sola baine-na *t_{itiŋ bulaŋ}*
- that month come 3ABS NAME with wife-3GEN *t_{that month}*
- ‘That month, Ramang came with his wife.’
- b. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
- naparákke^H i _____ waláo^H gája
- na-parakke? i *t_{gaja}* balao gad̪ʒa
- 3ERG-frighten 3ABS *t_{elephant}* mouse elephant
- ‘The elephant frightened the mouse.’

3.2 Building up to Crossing Constraints

The situation is different on two counts with a second class of movements in the language. The simplest steps within this class are a series of processes that can be collectively characterized as types of quantifier float: they are processes that draw quantificational elements out of DPs and raise them to higher positions along the clausal spine. The following examples introduce some of the quantificational elements of this class: they include *le'ba* 'exactly,' *tappa* 'just,' *to* 'too,' *tia* 'even,' *nasang* 'every,' *pira* 'some,' and *sangga* 'only.'

(35) *Mandar Quantifiers*

- a. Mau na-bengang [_{DP} sallessorang **le'ba'**], merau i pale'.
 though 3ERG-given a thousand exactly request 3ABS more
 'Though given exactly a thousand, he asked for more.'
- b. Mau na-olo'i [_{DP} **sangga'** kindo'-u], minassa i seni-u macoa.
 though 3ERG-like only mom-1GEN clear 3ABS art-1GEN good
 'Though appreciated by only my mom, my art is clearly good.'

The quantificational elements in this system can be seen in their underlying adnominal positions in two contexts: in fragments and in the smallest class of non-finite clauses in the language. The preceding examples show this fact in one environment of the second type: in the bare *voiceP* complements to the preposition *mau* ‘though.’ As more and more functional structure is added to the clausal spine, however, these elements begin to move. The following examples illustrate the first step in this system: Mandar has a class of aspectual auxiliaries that surface above the verb, and when any of these elements are introduced into the non-finite clauses above, these quantifiers are forced to move left. The quantifiers that follow their associates when adnominal, like *le’ba* ‘exactly,’ and *nasang*

‘every,’ raise to positions that follow the verb (36a); the quantifier *sangga* ‘only,’ which precedes its associate when adnominal, raises to a position that precedes the verb (36b).

(36) *Quantifiers to the Left*

- a. Mau **rua** na-bengang **le’ba’** [_{DP} sallessorang ____], merau dua i.
 though once 3ERG-given exactly a thousand request still 3ABS
 ‘Though once given exactly a thousand, he still asks.’
- b. Mau **rua** **sangga’** na-olo’i [_{DP} ____ kindo’-u], laku i seni-u.
 though once only 3ERG-like mom-1GEN sell 3ABS art-1GEN
 ‘Though once appreciated by only my mom, my art sells.’

The situation is slightly different in the finite clauses of the language, which are distinguished by the appearance of absolutive agreement. The enclitic that spells out absolutive agreement is a second-position element, and in the clauses that host overt auxiliaries, it follows the highest. In the clauses that contain this agreement, all of the quantifiers must undergo this first step of movement: thus *le’ba* ‘exactly’ will raise to follow the verb (37a) and *sangga* ‘only’ will raise to precede it (37b).

(37) *Finite Clauses: Obligatory Movement*

- a. Rua i na-bengang **le’ba’** [_{DP} sallessorang ____].
 once 3ABS 3ERG-give exactly a thousand
 ‘She once gave him exactly a thousand.’
- b. Rua i **sangga’** na-bengang [_{DP} ____ sallessorang].
 once 3ABS only 3ERG-give a thousand
 ‘She once gave him only a thousand.’

In finite clauses, however, a second step of raising targets five of these elements: *tappa* ‘just,’ *to* ‘too,’ *tia* ‘even,’ *nasang* ‘every,’ and *pira* ‘some.’ The following examples illustrate with *nasang* ‘every:’ it appears adnominally in non-finite clauses without auxiliaries (38a), raises to follow the verb in the non-finite clauses that host auxiliaries (38b), and, in the finite clauses that host absolutive agreement, raises to follow the highest auxiliary (38c).

(38) *The Second Step*

- a. Mau na-ita [_{DP} polisi **nasang**], na-cowa dua i maindong.
 though 3ERG-see police every 3ERG-try still 3ABS run

‘Though seen by every policeman, he still tried to run.’

- b. Mau **rua** na-ita **nasang** [_{DP} polisi ____], maccoro’ dua i.
 though once 3ERG-see every police steal still 3ABS

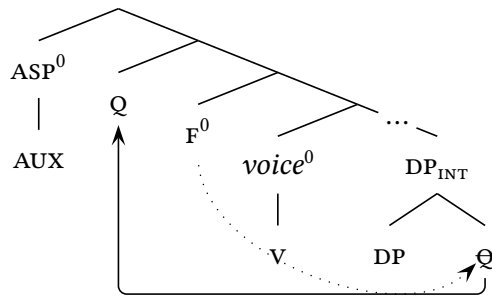
‘Though once seen by every policeman, he still steals.’

- c. Rua **nasang** i na-ita ____ [_{DP} polisi ____].
 Once every 3ABS 3ERG-see police

‘Once every policeman saw him.’

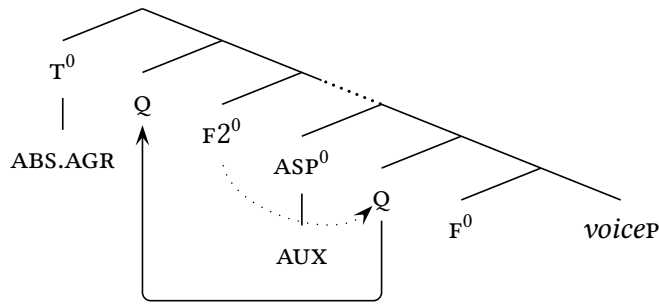
To capture these effects, I propose that there are two heads on the clausal spine that attract the quantifiers in this system. The first sits just above the *voiceP*, absent in the bare *voiceP*-complements of (35); it appears in all clauses that host auxiliaries, and as a result, I propose that it falls just beneath the head that hosts the lowest AUX (ASP⁰; on the syntax of the auxiliary system: (Brodkin, 2022a)). Its role is to trigger the first step of movement, which draws all of the quantifiers to a single syntactic position: the juncture between the *voiceP* and the middlefield (39). From this position, I assume that these quantifiers are linearized in slightly different ways in the postsyntax: based on the same item-specific preferences that guide their linearization in the DP, the quantifier *sangga*’ is forced to fall before the verb, while the remaining elements in the system are forced to fall after it.

(39) *Quantifier Float: Step One*



Higher up along the clausal spine, I propose that there is a second head that attracts a subset of the quantifiers to a higher position. Given that this head emerges only in finite clauses, I propose that it falls above all the elements in the auxiliary system and just below τ^0 : the source of absolute agreement (Brodkin, 2022b). I assume that the quantifiers attracted to this position are once again linearized in the postsyntax: just like the absolute agreement enclitics, they are repositioned to follow the highest auxiliary.

(40) *Quantifier Float: Step Two*



Before moving forward, we can note that there is a context where this second step has no visible linear effect: in finite verb-initial clauses, the quantifiers that raise to this higher position simply follow the verb (41a). In this syntactic environment, then, they take the same linear position as *le'ba* 'exactly' (41b), which syntactically fails to undergo this higher step. In finite clauses, then, these quantifiers can be said to show a kind of 'second-position' behavior: like absolutive agreement, they follow the verb or the highest auxiliary. Despite this fact, I assume that the syntax in these two configurations is essentially the same: these quantifiers undergo both steps of movement in finite verb-initial clauses. To capture their linear position, then, I assume that they—like *le'ba*—are positioned after the highest following auxiliary or verb in the postsyntax.

(41) *Finite Verb-Initial Clauses: Second Step Obscured*

- a. Na-ita **nasang** i [_{DP} polisi ____].
 3ERG-see 3ABS police
 'Every policeman saw him.'
- b. Na-bengang **le'ba'** i [_{DP} sallessorang ____].
 3ERG-give exactly 3ABS a thousand
 'She gave him exactly a thousand.'

3.3 Crossing Constraints

With this much in place, we can return to the phonology. The quantifiers in this system invariably form independent ω s, and in all positions, they are parsed into minimal ϕ s with their hosts. When there is no movement involved, they have no special effect on the prosody of the clause. When quantifiers are introduced into adnominal positions,

then, they trigger nothing more than the regular prosodic shifts created by introducing additional words into the DP. We can see this symmetry by comparing these elements to adjectives, which are also typically postnominal and which also form ω s that are parsed into minimal ϕ s with their preceding nouns. When an adjective is introduced into the object in a vSO clause, creating a four-word string, the ensuing parse must be $\{\phi_{[MAX]} \text{ vs-}\} \{\phi_{[MAX]} \text{ O-ADJ-}\}$ (42a). When an adnominal quantifier is introduced in the same position, in turn, the parse is once again $\{\phi_{[MAX]} \text{ vs-}\} \{\phi_{[MAX]} \text{ O-Q-}\}$ (42b).

(42) *Quantifiers without Movement: No Special Prosody*

- a. $\{\phi_{[MAX]} \quad \quad \quad \}$ $\{\phi_{[MAX]} \quad \quad \quad \}$ $\{\phi_{[MAX]} \quad \quad \quad \}$
- | | | | | | |
|--------|-------------------|----------------------|---------|----------------------|--------------------------|
| mo | néte ^H | iripáʔi ^H | gúru | malɪŋáo ^H | álo sattu ^H , |
| mau | na-itai | iripaʔi | guru | malɪŋao | allo sattu |
| though | 3ERG-seek | NAME | teacher | tall | day saturday |
- ‘Though Ripa’i looked for [–FIN] the tall teachers on saturday,’
- b. $\{\phi_{[MAX]} \quad \quad \quad \}$ $\{\phi_{[MAX]} \quad \quad \quad \}$ $\{\phi_{[MAX]} \quad \quad \quad \}$
- | | | | | | |
|--------|-------------------|----------------------|---------|--------------------|--------------------------|
| mo | néte ^H | iripáʔi ^H | gúru | násan ^H | álo sattu ^H , |
| mau | na-itai | iripaʔi | guru | nasan | allo sattu |
| though | 3ERG-seek | NAME | teacher | all | day saturday |
- ‘Though Ripa’i looked for [–FIN] all the teachers on saturday,’

The situation is different when the quantifiers move. Each type of quantifier float triggers the same phonological effect as WH-movement: its path must be wrapped by a ϕ . The following examples illustrate with *nasang* ‘every,’ focusing on finite clauses of the order vSOD. When *nasang* originates in the O, it raises to follow the verb and forces the exceptional prosody $\{\phi_{[MAX]} \text{ v Q s O } t_Q \} \{\phi_{[MAX]} \text{ D } \}$ —implicating an overheavy four-word ϕ that stretches from the verb-quantifier string across the s and o to its trace (43a). When *nasang* originates in the D, in the same vein, it raises and forces the even more exceptional prosody of $\{\phi_{[MAX]} \text{ v Q s O D } t_Q \}$ —featuring an even heavier five-word ϕ (43b).

(43) *Quantifier Float: Path = ϕ*

- a. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
- | | | | | | |
|------------|---------------------|---------------------|-------------------|--------------|-------------------|
| nawawáŋan | nása ^H ĩ | sanéke ^H | wúku ^H | _____ | gúru ^H |
| na-babaŋaŋ | naŋaŋ i | sanaeke | buku | t_{nasang} | guru |
| 3ERG-bring | all | 3ABS | kid | book | teacher |
- ‘The kids brought all the books to the teacher.’
- b. $\{\phi_{[MAX]}\}$ $\}$
- | | | | | | |
|------------|---------------------|---------------------|-------------------|-------------------|--------------|
| nawawáŋan | nása ^H ĩ | sanéke ^H | wúku ^H | yúru ^H | _____ |
| na-babaŋaŋ | naŋaŋ i | sanaeke | buku | guru | t_{nasang} |
| 3ERG-bring | all | 3ABS | kid | book | teacher |
- ‘The kids brought books to all the teachers.’

To provide another example, the facts are identical with *sangga* ‘only.’ This element raises to precede the verb in finite verb-initial clauses of the order *vsOD*, undergoing only the first step of movement in tree (39). This step of movement also forces the construction of a ϕ around its path: here, stretching from *sangga*’ to the right edge of its associate (extending slightly beyond the trace of *sangga*’, plausibly because the trace must be parsed into a ϕ with the rest of the associate). The following examples illustrate: when *sangga*’ raises out of an *o*, the parse is $\{\phi_{[MAX]} Q v s t_Q o\} \{\phi_{[MAX]} D\}$ (44a). When *sangga*’ raises out of a *D*, the parse must be $\{\phi_{[MAX]} Q v s o t_Q D\}$ (44b).

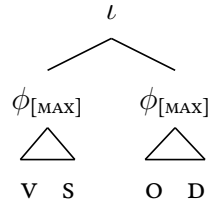
(44) *Second Example: Only-Float*

- a. $\{\phi_{[MAX]}\}$ $\{\phi_{[MAX]}\}$
- | | | | | | |
|--------|-------------------------|---------------------|---------------|-------------------|-------------------|
| sáŋga? | nawawáŋa ^H ĩ | sanéke ^H | _____ | wúku ^H | gúru ^H |
| sáŋga? | na-babaŋaŋ i | sanaeke | $t_{sangga?}$ | buku | guru |
| only | 3ERG-bring | 3ABS | kid | book | teacher |
- ‘The kids brought only books to the teacher.’
- b. $\{\phi_{[MAX]}\}$ $\}$
- | | | | | | |
|--------|-------------------------|---------------------|-------------------|---------------|-------------------|
| sáŋga? | nawawáŋa ^H ĩ | sanéke ^H | wúku ^H | _____ | yúru ^H |
| sáŋga? | na-babaŋaŋ i | sanaeke | buku | $t_{sangga?}$ | guru |
| only | 3ERG-bring | 3ABS | kid | book | teacher |
- ‘The kids brought books to only the teacher.’

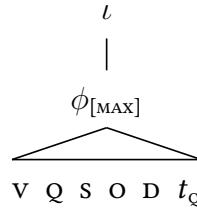
We can thus conclude that quantifier float and only-float affect prosodic phrasing in the same way as *WH*-movement: they require ϕ s to be built around their paths. When these processes escape the *D* in a four-word *vsOD* clause, to illustrate, they trigger a shift from the parse $\{\phi_{[MAX]} vs\} \{\phi_{[MAX]} OD\}$ (45a) to the parse $\{\phi_{[MAX]} v s o D\}$ (45b)-(45c).

(45) *Wrap-Chain: Two More Operations*

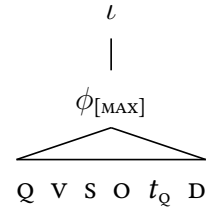
a. Regular Parse



b. Quantifier Float



c. Only-Float



We can now observe that the Wrapping Effect correlates with the Crossing Constraint. The following examples show that quantifier float cannot cross the path of WH-movement. In a VSOD clause where WH-movement targets the O, quantifier float can proceed inside this path from the S to the V (46a), but not across this path from the D to the V (46b).

(46) *Quantifier Float: the Crossing Constraint*

- a. WH V Q S t_{QF} t_{WH} D
A mu-sanga na-bengang nasang sola-mu t_{nasang} t_{apa} guru?
what 2ERG-think 3ERG-give all friend-2GEN t_{all} t_{what} teacher
‘What do you think all your friends gave to the teacher?’
- b. WH V Q S t_{WH} D t_{QF}
*A mu-sanga na-bengang nasang sola-mu t_{apa} guru t_{nasang}?
what 2ERG-think 3ERG-give all friend-2GEN t_{what} teacher t_{all}
INTENDED: ‘What do you think your friends gave to all the teachers?’

The following examples show that the same constraint holds over only-float: when a WH-phrase moves from the position of the O in a VSOD clause, only-float can proceed inside this path from the S to the V (47a), but not across this path from the D to the V (47b).

(47) *Only-Float: the Crossing Constraint*

- a. WH Q V t_{QF} S t_{WH} D
A sangga’ na-bengang t_{sangga’} sola-mu t_{apa} guru?
what only 3ERG-give t_{only} friend-2GEN t_{what} teacher
‘What did only your friends give to the teacher?’
- b. WH Q V S t_{WH} t_{QF} D
*A sangga’ na-bengang sola-mu t_{apa} t_{sangga’} guru?
what only 3ERG-give friend-2GEN t_{what} t_{only} teacher
INTENDED: ‘What did your friends give to only the teacher?’

4 Deriving Path Containment

The correlation with Wrapping Effects suggests that the Mandar Crossing Constraints are grounded in the phonological signatures of the movements in this system. Restricting our attention to the phonologically active movements, we then arrive at two questions:

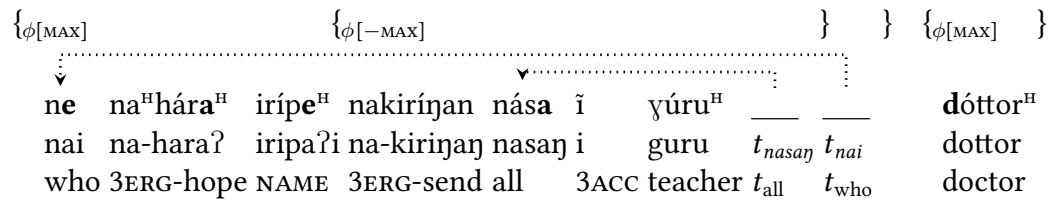
(48) *Guiding Questions: the Phonological Account*

- What property allows the paths of phonologically active movements to nest?
- What property prevents phonologically active movements from crossing?

4.1 Nested Paths: Prosodic Recursion

When phonologically active movements are nested, our diagnostics for the maximal ϕ show that a ϕ is built around the path of the outer movement. The same diagnostics do not allow us to detect the presence of a ϕ around the path of the inner movement. From a technical perspective, this is expected: if a second ϕ were built around the inner path, it would be contained in the ϕ of the outer path and would thus not be *maximal*. But from a theoretical perspective, this lack of evidence raises a question: when phonologically active movements are nested, are both paths still wrapped by ϕ s?

(49) *Inner Path* $\rightarrow \phi_{[-MAX]}$?



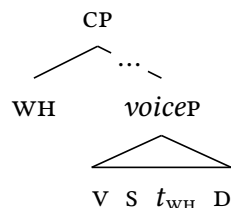
‘Who does Ripa’i hope all the teachers sent to the doctor?’

To answer this question, we can revisit the phonology of the Wrapping Effect. Up to this point, we have only seen examples where the paths of movement have always been wrapped by maximal ϕ s. But there is a phonological reason that these ϕ s have been maximal: the paths of movement, in all of these examples, have always contained three or more ω s. The following diagrams illustrate, starting from a VSOD clauses where WH-movement targets the o (50a). In this context, WRAP(CHAIN) should demand the construction of a ϕ around the string WH-VS- t_{WH} . The isomorphism constraints should then demand the construction of a higher ϕ around the string WH-VS- t_{WH} -D, as this string forms an XP (50b).

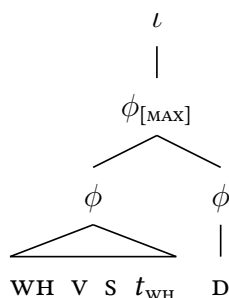
But the construction of this higher ϕ should be blocked by the Ternarity Constraint, which bans ϕ s of more than three ω s. In other words, the Ternarity Constraint blocks the construction of higher ϕ s around the ϕ that wraps the path of WH-movement—and as result, the wrapping ϕ becomes maximal (50c).

(50) *Wrap-Chain and the Ternarity Constraint*

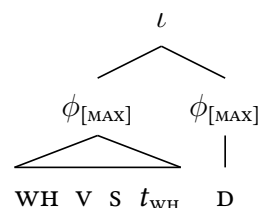
a. Syntax



b. Isomorphic Parse



c. Ternarity Constraint



We can sidestep this problem by turning to clauses where WH-movement moves along shorter paths. In that context, we see something different. In a vSO clause where WH-movement targets the s, for instance, our diagnostics reveal that a maximal ϕ is built around the full WH v- t_{WH} -O string, rather than just the narrow path of WH-movement (51a). In a vOD clause where WH-movement targets the O, in the same vein, our diagnostics show that a maximal ϕ surrounds the full string WH v- t_{WH} -O, not just the WH-path (51b). What these facts suggest is the following: when the path of WH-movement contains only two ω s, it is not always visibly parsed into a maximal ϕ .

(51) *Binary Paths: Path $\neq \phi_{MAX}$*

a. $\{\phi_{MAX}\}$

né	na ^H máli ^H		wúŋa ^H
nai	na-m-alli	t_{nai}	buŋa
who	will-ANTIP-buy	t_{who}	flower

‘Who will buy flowers?’

b. $\{\phi_{MAX}\}$

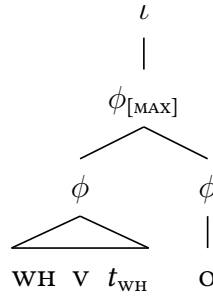
né	na ^H nakiriŋa ^H	ĩ	yúru ^H
nai	na-na-kiriŋaŋ	i	t_{nai}
who	will-3ERG-send	3ACC t_{who}	guru

‘Who will she send to the teacher?’

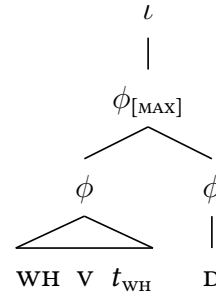
There is a natural way to understand these effects in terms of prosodic recursion. When the path of WH-movement contains two ω s, I propose that it is subjected to WRAP(CHAIN) and mapped to a ϕ . At the same time, I argue the syntax-prosody isomorphism constraints force the construction of a higher ϕ that contains the WH-v string and all other arguments that follow the v. The following diagrams show the result in clauses where this parse is not derailed by the Ternarity Constraint: the path of WH-movement is parsed into a non-maximal ϕ which is then nested within a maximal ϕ that corresponds to the CP.

(52) *Wrap-Chain* $\rightarrow$ *Non-Maximal Phrases*

a. VSO, s=WH

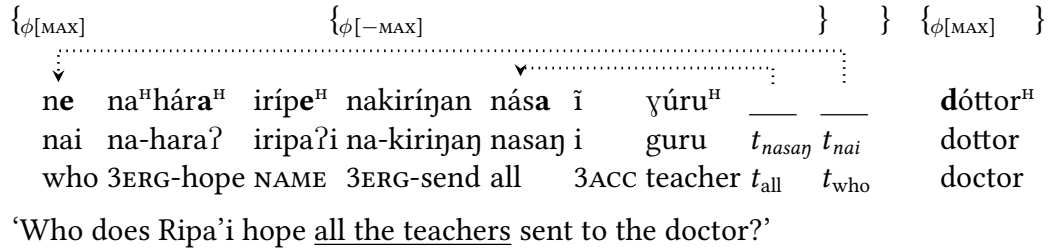


b. VOD, O=WH



This analytical path leads us to conclude that the phonologically active movements can satisfy WRAP(CHAIN) by forming phonotactically invisible ϕ s that are recursively embedded within larger ϕ s. This result, in turn, opens up a new perspective on nested paths. In the cases where quantifier float is nested in the path of WH-movement, I propose the recursive parse in (53): the inner path of quantifier float is parsed into a non-maximal ϕ that is embedded within the maximal ϕ that wraps the outer path of WH-movement.

(53) *Inner Path* $\rightarrow \phi_{[-MAX]}$



The result is an account that derives the acceptable status of nested paths from an architectural principle of prosodic constituency: the movements in this system must be wrapped by ϕ s, and it is possible for the ϕ to be recursively embedded (Itô & Mester 2007, 2012, 2013; Elfner 2012, 2015; Elordieta 2015; also Ladd 1986; Wagner 2005, 2010).

4.2 Crossing Paths: Proper Bracketing

We can now turn to the Crossing Constraint. The following example shows the prosody that should emerge in the contexts where phonologically active movements cross paths: each path should be wrapped by a ϕ and the two ϕ s should intersect.



INTENDED: 'What did your friend give to only the teacher?'

The result is a configuration which violates a foundational principle of constituency: the PROPER BRACKETING CONDITION of Nespor & Vogel 1986. This condition, presented in (55), forbids constituents from having multiple mother nodes. Its narrow force in this context is to ban configurations in which ϕ s intersect, placing the constituents beneath them in configurations where they are immediately dominated by more than one ϕ .

(55) *The Proper Bracketing Condition*

A unit of a given level of the [AU: prosodic] hierarchy is exhaustively contained in the superordinate unit of which it is a part.

In light of this violation, I propose that it is this constraint on prosodic constituency that gives rise to the Crossing Constraint. On this view, there is no issue in the syntax when steps of movement cross paths—a natural result on the hypothesis that representational constraints and linear order are entirely absent from that module. The problem emerges downstream instead: within the phonology, where specific movement dependencies are parsed into prosodic constituents, the prohibitions on crossing paths emerge from the architectural requirements imposed on prosodic constituents in the global phonological evaluation where prosodic structure is built

Stepping back from the particulars, then, we thus arrive at an analysis that derives the core effects in Mandar in a precise and elegant way: PATHS and representational constraints are eliminated from the syntax, the movements that cannot cross paths are linked by a transparent phonological generalization, the license for nesting emerges from the possibility for prosodic recursion, and the Crossing Constraint emerges from an architectural principle of constituency.

4.3 Repairing Path Containment: More Phonology

We can now leverage this understanding to make sense of the ways in which the Crossing Constraints in this system are resolved. When quantifier float and only-float cross the path of WH-movement in Mandarin, we have seen that the result is ungrammatical. But there is a way in which these crossing derivations are systematically rescued: *nasang* and *sangga* are exceptionally forced to remain adnominal when their movement would violate the Crossing Constraint. Thus in VSOD clauses where WH-movement targets the O, *nasang* and *sangga* move when they originate in the S (46a)-(47a) but remain adnominal when they originate in the D. In this context, they take the same positions that they take in non-finite contexts: *nasang* follows its associate (56a) and *sangga* precedes its own (56b).

(56) *Repair to the Crossing Constraint: Suspending Quantifier Float*

- a. WH V V Q S t_{WH} D Q
 A mu-sanga na-bengang ____ sola-mu t_{apa} guru **nasang**?
 what 2ERG-think 3ERG-give all friend-2GEN t_{what} teacher all
 ‘What do you think your friends gave to all the teachers?’
- b. WH Q V S t_{WH} Q D
 A ____ na-bengang sola-mu t_{apa} **sangga**’ guru?
 what only 3ERG-give friend-2GEN t_{what} only teacher
 ‘What did your friends give to only the teacher?’

We can understand this effect with a brief foray into the syntax of scope. In the non-finite clauses in which *nasang* and *sangga* always remain adnominal, it is possible to introduce the clausal negator *ndang*. This element surfaces above all auxiliaries in the middle field, and Brodtkin 2022a argues that it sits in POL⁰. What is important at present is the following: in the non-finite contexts where *nasang* and *sangga* remain adnominal, they must take scope beneath negation. The following example illustrates with *nasang*.

(57) *Nasang: Scopes Beneath Negation in Non-Finite Clauses*

- a. Mau [PolP **ndang** [voiceP na-ita polisi **nasang**], maindong dua i.
 though not 3ERG-see police all run still 3ABS
 ‘Though not seen by all the police, he still ran.’
- b. POSSIBLE: NOT > ALL (“some policemen saw him; others did not”)
- c. IMPOSSIBLE: ALL > NOT (“no policemen saw him”)

In the finite clauses that force movement of *nasang*, two patterns of scope are possible. When *nasang* raises to its usual high position, it must take scope over negation (58a). But *nasang* can also remain in its intermediate postverbal position in finite clauses that contain negation, and in this context, it takes scope beneath negation (58b).

(58) *Nasang: Movement and Scope*

- a. [_{PolP} **Ndang nasang** i [_{voiceP} na-ita polisi *t_{nasang}* iAli]].
 not all 3ABS 3ERG-see police *t_{all}* NAME
 ALL > NOT: ‘All the policeman were such that they failed to see Ali.’
- b. [_{PolP} **Ndang** i [_{voiceP} na-ita **nasang** polisi *t_{nasang}* iAli]].
 not 3ABS 3ERG-see all police *t_{all}* NAME
 NOT > ALL: ‘It was not the case that all the policeman saw Ali.’

The force of these observations is to suggest that there is a close link in this system between syntactic position and scope: *nasang* must scope below negation in the non-finite contexts where it remains adnominal in the syntax (57), and it can only take scope above negation when it moves to its highest syntactic position (58a)—a step that is suspended, in an interesting way, when *nasang* scopes below negation in a finite clause.

It is this relationship between syntactic movement and scope that will allow us to understand the nature of our repair to the Crossing Constraint. In the contexts where *nasang* is forced to remain adnominal in a clause that launches WH-movement, this element is not forced to take low scope. Instead, in the clauses where its movement is seemingly blocked by the Crossing Constraint, *nasang* can freely scope above or below the negator (59).

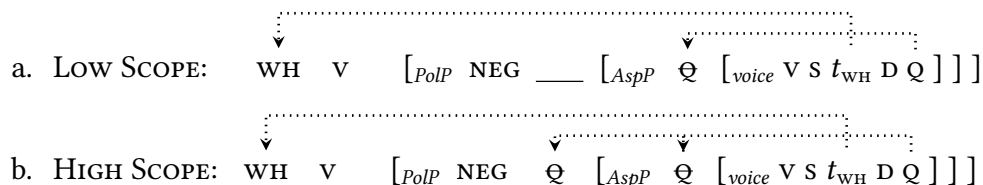
(59) *Nasang: Flexible Scope in Crossing Contexts*

- a. WH V NEG Q V S *t_{WH}* D Q
 A mu-sanga ndang ____ na-bengang sola-mu *t_{apa}* guru **nasang**?
 what 2ERG-think not all 3ERG-give friend-2GEN *t_{what}* teacher all
 ‘What do you think your friends didn’t give to all the teachers?’
- b. POSSIBLE: NOT > ALL (“some teachers got a gift, others did not”)
- c. POSSIBLE: ALL > NOT (“no teacher got a gift”)

What this fact suggests, I believe, is that *nasang* moves in a syntactically regular way in the contexts where it seems to be held in place by the Crossing Constraint. The repair that emerges in that context, then, is fully oriented toward the surface: the phonology cannot properly prosodify the path of overt quantifier float, since it would be forced to

build a ϕ that violated the Proper Bracketing Condition, and as a result, it forces the realization of *nasang* in its base position—in response to an output constraint, in other words, it forces this step of movement to be covert. The following diagrams illustrate this analysis, marking covert movement by striking out higher copies. When *nasang* scopes beneath negation in this context (59b), it raises covertly to the intermediate position (60a); when *nasang* scopes above negation (59c), it raises covertly to the higher one (60b).

(60) *Crossing Constraint* $\rightarrow$ *Covert Quantifier Float*



We thus arrive at an account that fits neatly with the mainstream Minimalist theory of syntax-phonology interaction. In the syntax, quantifier float must operate undisturbed in the contexts where it violates the Crossing Constraint—a natural result if phonological information is absent from that module (Zwicky & Pullum, 1986). At the derivational moment where the syntax and phonology meet, the phonology must then recognize that spelling out *nasang* in its high position would force violations of *Wrap-Chain* or the Proper Bracketing Condition. The phonology must then force *nasang* to be spelled out in a lower position, yielding a case of prosodically driven lower-copy spell out (Bošković, 2001).

5 Conclusion

This paper has argued that Path Containment Effects in Mandar must be understood to emerge from constraints on output prosodic structure: the movements that cannot cross are those that require phonological phrases to be built around their paths, and these movements cannot cross paths because phonological phrases cannot intersect.

To the extent that this analysis is successful for Mandar, it lays the foundations to reconsider similar phenomena in phonological terms. The most natural targets would be other Path Containment Effects: for instance, the restrictions that arise around crossing $\bar{A}$ -dependencies in English. I believe that the English phenomena are also grounded in conflicts over prosodic constituency, and the beginnings of the case for this view emerge from a well-known phonological contrast between A-movement, which can cross the paths of $\bar{A}$ -dependencies, and WH-movement, which cannot. In English, the τ^0 *to* can take a re-

duced allomorph and encliticize to a preceding verb across the trace of raising (61a) but not across the trace of WH-movement (61b) (Lightfoot, 1976; Jaeggli, 1980).

- (61) a. $\{\phi_{\downarrow} \dots \dots \dots \}$
 I need *t* **a/*to** visit Sulawesi.
- b. $\{\phi_{\downarrow} \dots \dots \dots \} \{\phi_{[MAX]} \dots \dots \dots \}$
 Who do you need *t* **to/*a** visit Sulawesi?

There is a natural way to understand this restriction in output-oriented terms: *to* cannot encliticize across the trace of WH-movement because the path of WH-movement—in English as in Mandar—must also be wrapped by a phonological phrase.

Pushing further with this view, the same type of output-oriented logic may then be leveraged to explain specific island constraints and freezing effects. It now seems likely that the known network of island constraints and freezing effects emerges from the combined work of many different systems, including constraints in the syntax and semantics alongside pressures in the system of performance. But there are several additional island effects that seem particularly amenable to reanalysis in output phonological terms. One example is the Adjunct Island Condition (Ross, 1967). It has been repeatedly noted in different languages and different theoretical contexts that adjuncts fail to be parsed into tight prosodic domains with their hosts (Chen, 1987; Hale & Selkirk, 1987; Truckenbrodt, 1999). It thus seems natural to imagine that certain adjuncts will form prosodic islands for the movement operations that whose paths must be parsed into phonological phrases. A case in point involves English adjunct CPs that are headed by *if*. It is ungrammatical to extract from these adjunct CPs when they form independent intonational phrases (62b)—a parse that can be seen in some varieties of American English by the possibility of ejectivizing a final voiceless stop at the right edge of the *ι*. But I find it fully acceptable to extract from adjunct CPs of this type when they are parsed into a single intonational phrase with the matrix clause (62c)—a prosodic configuration that should allow for the construction of a phonological phrase around the path of WH-movement.

- (62) a. Wait, wait, tell me one more time...
- b. $\{\iota_{\downarrow} \dots \dots \dots \} \dots \dots \{\iota_{\downarrow} \dots \dots \dots \}$
 *What would you get sick [sik'] if you ate *t*?
- c. $\{\iota_{\downarrow} \dots \dots \dots \}$
 What would you get sick [sik] if you ate *t*?

Turning now to the phonology, our results provide further evidence that the syntax-prosody interface can mobilize prosodic phrasing to preserve fine syntactic relationships—and in particular, that it can mandate the construction of phonological phrases from the heads to the tails of movement chains. As a result, our results raise questions about the types of syntactic relationships that can be prosodified in this way—and in particular, whether similar prosodic effects might be forced to preserve the relationships argued to link c^0 to in-situ WH-phrases with c^0 s in Japanese (Deguchi & Kitagawa, 2002; Ishihara, 2002, 2003, 2005; Hirotsu, 2005; Smith, 2005, 2013, 2014) (and cf. Richards 2010, 2016; Branen 2018).

In search of an explanation for why this prosody emerges around WH-movement, at last, a final possibility rises to the fore. There is a prosodic divide between two types of $\bar{A}$ -movement in Mandarin: WH-movement requires wrapping but scrambling and topicalization do not. This division, interestingly, correlates with a second divide: WH-movement can target non-referential elements (63) but scrambling and topicalization cannot (64). The $\bar{A}$ -movements that are not wrapped, then, are those that only target definite DPs.

(63) *Wh-Movement: Can Target Non-Referential Constituents*

$\{\phi_{[MAX]} \dots \dots \dots \}$
 $\downarrow$
 sápa^H wémme^H _____
 saʔapa bemme $t_{saʔapa}$
 how many fall $t_{how\ many}$
 ‘How many fell?’

(64) *Scrambling/Topicalization: Only Definite Constituents*

- $\dots \dots \dots \downarrow$
 a. *Mam-baca a’ t_{buku} dionging buku
 ANTIP-read 1ABS $t_a\ book$ yesterday a book
 INTENDED: ‘I read yesterday a book.’
 $\downarrow$
 b. *Buku, mam-baca a’ t_{buku} dionging
 a book ANTIP-read 1ABS $t_a\ book$ yesterday
 INTENDED: ‘A book, I read yesterday.’

From this perspective, it seems tempting to link this prosodic split to the differences in the identity of the element at the base of the movement chain: scrambling and topicalization, more specifically, may leave null pronominal elements at the tails of their chains (Cinque, 1990; Postal, 1994, 1998). If such an account turned out to be correct, it may then be possible to derive the distribution of wrapping effects from an interface requirement that held over only a specific subset of null elements at the bases of chains—and thus open up a phonological path to understand the island-ameliorating effects of resumption.

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